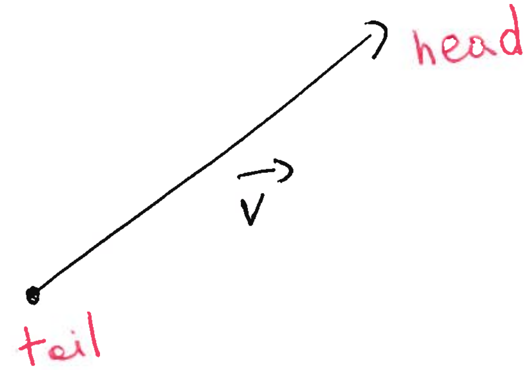
11.2 Vectors

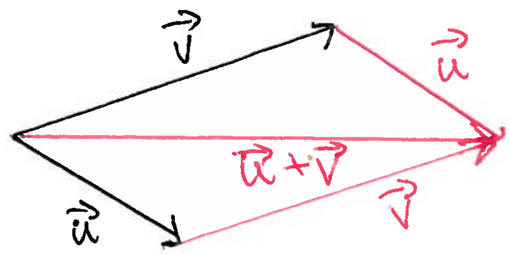
Vectors are determined by their <u>magnitude</u> & <u>direction</u>



$\|\vec{v}\|$ is the magnitude

To <u>add vectors</u> $\vec{u}$ & $\vec{v}$

place the tail of $\vec{u}$ on the head of $\vec{v}$

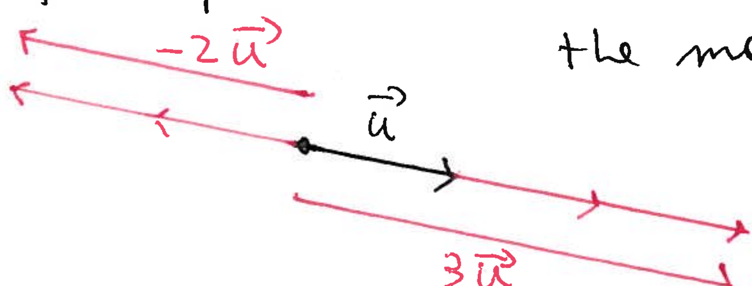


$$\vec{u}+\vec{v}=\vec{v}+\vec{u}$$

$$(\vec{u}+\vec{v})+\vec{w}=\vec{u}+(\vec{v}+\vec{w})$$

$3\vec{u}$ is a vector with the same direction as $\vec{u}$ but 3 times the magnitude

$-2\vec{u}$ has the oposite direction and 2 times the magnitude



$\lambda \in \mathbb{R}_{>0}$

$\lambda\vec{u}$ is a vector with the same direction as $\vec{u}$ but the magnitude is λ times that of $\vec{u}$

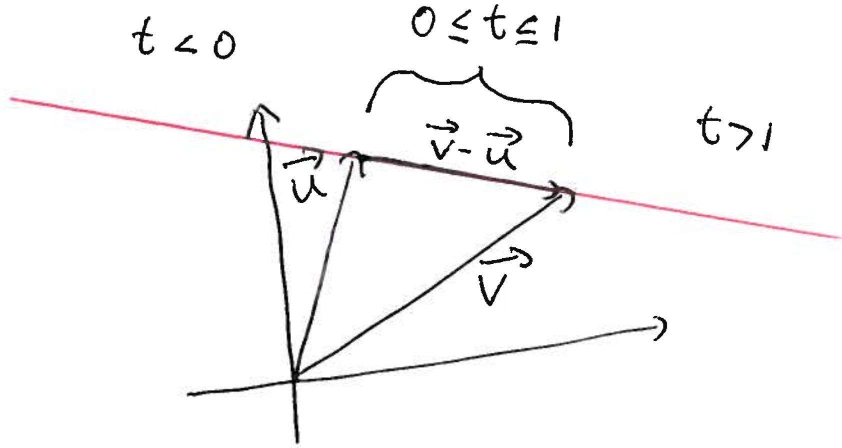
$\vec{0}$ is the vector of 0 magnitude

$\vec{0} + \vec{u} = \vec{u} = \vec{u} + \vec{0}$

Note that $-\vec{u} + \vec{u} = \vec{0} = \vec{u} + (-\vec{u})$

<u>Example</u> the points on a line through the end points of the vectors $\vec{u}$ & $\vec{v}$ are given by $\vec{u} + t(\vec{v} - \vec{u})$

or $\vec{u}(1-t) + t\vec{v}$



Line through $(1,0,1)$ & $(2,-1,-3)$

$(1,0,1)(1-t) + t(2,-1,-3) = (1+t, -t, 1-4t)$

$t=0 \rightarrow (1,0,1)$ $\qquad t=1 \rightarrow (2,-1,-3)$

$t=1/2 \rightarrow (3/2, -\frac{1}{2}, -1)$ mid point

$t=-2 \rightarrow (-1, 2, 9)$ etc.

It is easier to work algebraically

$\mathbb{R}^3$ $\vec{u} = (u_1, u_2, u_3)$ $\vec{v} = (v_1, v_2, v_3)$

$\vec{u} + \vec{v} = (u_1+v_1, u_2+v_2, u_3+v_3)$ $\vec{0} = (0,0,0)$

$\lambda \in \mathbb{R}$ $\lambda\vec{u} = (\lambda u_1, \lambda u_2, \lambda u_3)$ $-\vec{u} = (-u_1, -u_2, -u_3)$

$\|\vec{u}\| = \sqrt{u_1^2 + u_2^2 + u_3^2}$

$\|\lambda\vec{u}\| = \sqrt{(\lambda u_1)^2 + (\lambda u_2)^2 + (\lambda u_3)^2} = \sqrt{\lambda^2 \cdot (u_1^2 + u_2^2 + u_3^2)} = \sqrt{\lambda^2} \cdot \sqrt{u_1^2 + u_2^2 + u_3^2}$

$= |\lambda| \cdot \|\vec{u}\|$

$\vec{u}$ is a unit vector if $\|\vec{u}\| = 1$

$\vec{i} = (1,0,0)$ $\vec{j} = (0,1,0)$ $\vec{k} = (0,0,1)$ are the standard unit vectors

$\vec{u} = (u_1, u_2, u_3) = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$

Rules ① $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ ② $(\vec{u} + \vec{v}) + \vec{w} = u + (\vec{v} + \vec{w})$

③ $\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$ ④ $\vec{u} + (-\vec{u}) = 0$

⑤ $a(b\vec{u}) = (ab)\vec{u}$ ⑥ $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$

⑦ $(a+b)\vec{u} = a\vec{u} + b\vec{u}$ ⑧ $1 \cdot \vec{u} = \vec{u}$

⑨ $\|a\vec{u}\| = |a| \, \|u\|$

Proof of ⑦ $(a+b)\vec{u} = (a+b)(u_1, u_2, u_3) = ((a+b)u_1, (a+b)u_2, (a+b)u_3)$

$= (au_1 + bu_1, au_2 + bu_2, au_3 + bu_3) =$

$= (au_1, au_2, au_3) + (bu_1, bu_2, bu_3) = a\vec{u} + b\vec{u}$

Ex find unit vector in direction of $(2,3)$

$\|(2,3)\| = \sqrt{2^2+3^2} = \sqrt{13}$ $\qquad \frac{1}{\sqrt{13}}(2,3) = \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right)$

check! $\left\|\left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right)\right\| = \sqrt{\frac{4}{13}+\frac{9}{13}} = \sqrt{\frac{13}{13}} = 1$ ✓

HOMEWORK:

Ex Let $\vec{u} = (2,1,3)$ $\quad \vec{v} = (-1,4,-2)$. Compute

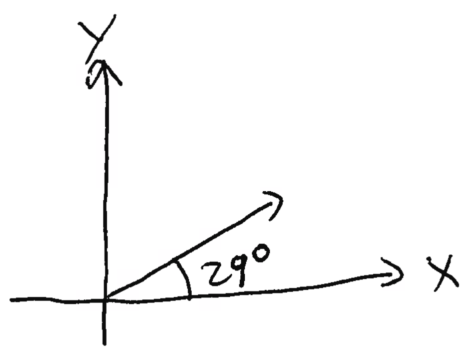
$\|\vec{u}\|$, $\|\vec{v}\|$, $\vec{u}+\vec{v}$, $\vec{u}-\vec{v}$, $3\vec{u}-2\vec{v}$.

- Find the unit vector in the direction of $\vec{u}$.

Ex Let $\vec{u} = (2,1)$ $\quad \vec{v} = (1,2)$. Draw

$\vec{u}+\vec{v}$, $\vec{u}-\vec{v}$, $2\vec{u}-4\vec{v}$.

Ex Use a calculator to compute the components of the vector $\vec{v}$:



$\|v\| = 3$